CLIMATE CLASSIFICATION (KOPPEN) / INDIA CLIMATIC REGIONS

CLASSIFICATION DESIGN GATE → KOPPEN MAJOR GROUPS → TROPICAL CODE GRAMMAR → DRY-CLIMATE GATE → C-D THRESHOLD FIREWALL → NORMAL-TO-TREND LADDER

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

◆ PRIMARY CORE FLOW ◆ SUBORDINATE EXTRA ◆ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artifacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • active generation g2 • explicit approval of this exact generation is required • all prior artifacts unchanged

00

STAGE 00

CLASSIFICATION DESIGN GATE

CLASSIFICATION DESIGN GATE · COMPARISON, EXPLANATION OR PLANNING · TEMPERATURE, RAIN OR WATER BALANCE · THRESHOLDS, PERIOD, SCALE, INTERPOLATION · USEFUL MODEL WITH EXPLICIT LIMITS

PURPOSE → comparison, explanation or planning

VARIABLES → temperature, rain or water balance → METHOD → thresholds, period, scale, interpolation → OUTPUT → useful model with explicit limits

CLASSIFICATION	DIAGNOSTIC BASIS	PROCESS LINK	EXAM TRAP
• PURPOSE → comparison, explanation or planning • VARIABLES → temperature, rain or water balance	• METHOD → thresholds, period, scale, interpolation • OUTPUT → useful model with explicit limits	• PURPOSE → comparison, explanation or planning • VARIABLES → temperature, rain or water balance	• METHOD → thresholds, period, scale, interpolation • OUTPUT → useful model with explicit limits

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: METHOD → thresholds, period, scale, interpolation; OUTPUT → useful model with explicit limits.

01

STAGE 01

KOPPEN MAJOR GROUPS

KOPPEN MAJOR GROUPS · TROPICAL THERMAL REGIME · DRYNESS THRESHOLD FIRST · COLDEST-MONTH CONVENTION MATTERS · WARMEST MONTH BELOW 10 C; H ADDED LATER

A → tropical thermal regime

B → dryness threshold first → C / D → coldest-month convention matters → E → warmest month below 10 C; H added later

CLASSIFICATION	DIAGNOSTIC BASIS	PROCESS LINK	EXAM TRAP
• A → tropical thermal regime • B → dryness threshold first	• C / D → coldest-month convention matters • E → warmest month below 10 C; H added later	• A → tropical thermal regime • B → dryness threshold first	• C / D → coldest-month convention matters • E → warmest month below 10 C; H added later

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: E → warmest month below 10 C; H added later.

02

STAGE 02

TROPICAL CODE GRAMMAR

TROPICAL CODE GRAMMAR · NO DRY SEASON · MONSOON-SHORT DRY SEASON LOGIC · DRY WINTER · DRY SUMMER · FIRST LETTER CLIMATE, SECOND RAINFALL

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
Af → no dry season	Aw / As → dry winter / dry summer	Af → no dry season
Am → monsoon short dry season logic	RULE → first letter climate, second rainfall	Am → monsoon short dry season logic

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
• Af → no dry season • Am → monsoon short dry season logic	• Aw / As → dry winter / dry summer • RULE → first letter climate, second rainfall	• Af → no dry season • Am → monsoon short dry season logic

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Aw / As → dry winter / dry summer.

03

STAGE 03

DRY-CLIMATE GATE

DRY-CLIMATE GATE · TEMPERATURE + RAIN SEASON · DRYNESS THRESHOLD · PRECIPITATION BELOW THRESHOLD · B GROUP · BW DESERT · LESS DRY · BS STEPPE; H/K THERMAL SUFFIX

TEMPERATURE + RAIN SEASON → dryness threshold

PRECIPITATION BELOW THRESHOLD → B group → DRIER → BW desert → LESS DRY → BS steppe; h/k thermal suffix

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
• TEMPERATURE + RAIN SEASON → dryness threshold • PRECIPITATION BELOW THRESHOLD → B group	• DRIER → BW desert • LESS DRY → BS steppe; h/k thermal suffix	• TEMPERATURE + RAIN SEASON → dryness threshold • PRECIPITATION BELOW THRESHOLD → B group

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LESS DRY → BS steppe; h/k thermal suffix.

04

STAGE 04

C-D THRESHOLD FIREWALL

C-D THRESHOLD FIREWALL · COLDEST MONTH · C/D DISCRIMINATOR · CONVENTION 1 · MINUS 3 C BOUNDARY · CONVENTION 2 · 0 C BOUNDARY · ANSWER RULE

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
COLDEST MONTH → C/D discriminator	CONVENTION 2 → 0 C boundary	COLDEST MONTH → C/D discriminator
CONVENTION 1 → minus 3 C boundary	ANSWER RULE → state convention before mapping	CONVENTION 1 → minus 3 C boundary

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
• COLDEST MONTH → C/D discriminator • CONVENTION 1 → minus 3 C boundary	• CONVENTION 2 → 0 C boundary • ANSWER RULE → state convention before mapping	• COLDEST MONTH → C/D discriminator • CONVENTION 1 → minus 3 C boundary

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CONVENTION 2 → 0 C boundary; ANSWER RULE → state convention before mapping.

05

STAGE 05

NORMAL-TO-TREND LADDER

NORMAL-TO-TREND LADDER · STATED 30-YEAR REFERENCE · OBSERVATION MINUS NORMAL · SPREAD AND RECURRENCE · MULTI-PERIOD DIRECTION WITH UNCERTAINTY

NORMAL → stated 30-year reference

ANOMALY → observation minus normal → VARIABILITY → spread and recurrence → TREND → multi-period direction with uncertainty

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
• NORMAL → stated 30-year reference • ANOMALY → observation minus normal	• VARIABILITY → spread and recurrence • TREND → multi-period direction with uncertainty	• NORMAL → stated 30-year reference • ANOMALY → observation minus normal

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VARIABILITY → spread and recurrence; TREND → multi-period direction with uncertainty.

06

STAGE 06

CLIMATE-BOUNDARY MODEL

CLIMATE-BOUNDARY MODEL · MEASURED VARIABLES · QUALITY CONTROL · COMPARABLE RECORDS · INTERPOLATION + THRESHOLDS · CLASS MAP · GROUND REALITY · TRANSITION, NOT WALL

STATIONS / GRIDS → measured variables → QUALITY CONTROL → comparable records → INTERPOLATION + THRESHOLDS → class map → GROUND REALITY → transition, not wall

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
• STATIONS / GRIDS → measured variables • QUALITY CONTROL → comparable records	• INTERPOLATION + THRESHOLDS → class map • GROUND REALITY → transition, not wall	• STATIONS / GRIDS → measured variables • QUALITY CONTROL → comparable records

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INTERPOLATION + THRESHOLDS → class map; GROUND REALITY → transition, not wall.

07

STAGE 07

INDIA CONTROL MATRIX

INDIA CONTROL MATRIX · LATITUDE + ALTITUDE · THERMAL STRUCTURE · UPLIFT, RAIN SHADOW, COLD BARRIER · SEA DISTANCE · CONTINENTAL RANGE · MONSOON + WDS + OCEAN · SEASONAL VARIABILITY

CLASSIFICATION	DIAGNOSTIC BASIS	PROCESS LINK	EXAM TRAP
LATITUDE + ALTITUDE → thermal structure	SEA DISTANCE → maritime / continental range	LATITUDE + ALTITUDE → thermal structure	SEA DISTANCE → maritime / continental range
RELIEF → uplift, rain shadow, cold barrier	MONSOON + WDs + OCEAN → seasonal variability	RELIEF → uplift, rain shadow, cold barrier	MONSOON + WDs + OCEAN → seasonal variability

CLASSIFICATION	DIAGNOSTIC BASIS	PROCESS LINK	EXAM TRAP
• LATITUDE + ALTITUDE → thermal structure • RELIEF → uplift, rain shadow, cold barrier	• SEA DISTANCE → maritime / continental range • MONSOON + WDs + OCEAN → seasonal variability	• LATITUDE + ALTITUDE → thermal structure • RELIEF → uplift, rain shadow, cold barrier	• SEA DISTANCE → maritime / continental range • MONSOON + WDs + OCEAN → seasonal variability

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MONSOON + WDs + OCEAN → seasonal variability.

08

STAGE 08

INDIA CODE-MAP BOUNDARY

INDIA CODE-MAP BOUNDARY · COMMON TYPES · TROPICAL, DRY, SUBTROPICAL, HIGHLAND · EXACT CODE · DEPENDS ON DATASET AND THRESHOLD · DEPENDS ON SCALE AND INTERPOLATION · CITE SOURCE, PERIOD AND CONVENTION

SYSTEM / DEFINITION	PROCESS / MECHANISM	SPATIAL / INDIA PATTERN	HAZARD / LIMIT / RESPONSE
COMMON TYPES → tropical, dry, subtropical, highland	BOUNDARY → depends on scale and interpolation	COMMON TYPES → tropical, dry, subtropical, highland	BOUNDARY → depends on scale and interpolation
EXACT CODE → depends on dataset and threshold	RULE → cite source, period and convention	EXACT CODE → depends on dataset and threshold	RULE → cite source, period and convention

SYSTEM / DEFINITION	PROCESS / MECHANISM	SPATIAL / INDIA PATTERN	HAZARD / LIMIT / RESPONSE
• COMMON TYPES → tropical, dry, subtropical, highland • EXACT CODE → depends on dataset and threshold	• BOUNDARY → depends on scale and interpolation • RULE → cite source, period and convention	• COMMON TYPES → tropical, dry, subtropical, highland • EXACT CODE → depends on dataset and threshold	• BOUNDARY → depends on scale and interpolation • RULE → cite source, period and convention

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EXACT CODE → depends on dataset and threshold.

09

STAGE 09

CLASSIFICATION COMPARISON

CLASSIFICATION COMPARISON · TEMPERATURE + PRECIPITATION THRESHOLDS · MODIFIED THERMAL ZONATION · WATER BALANCE + PET · MATCH METHOD TO QUESTION PURPOSE

CLASSIFICATION	DIAGNOSTIC BASIS	PROCESS LINK	EXAM TRAP
KOPPEN → temperature + precipitation thresholds	THORNTHWAITE → water balance + PET	KOPPEN → temperature + precipitation thresholds	THORNTHWAITE → water balance + PET
TREWARTHA → modified thermal zonation	CHOICE → match method to question purpose	TREWARTHA → modified thermal zonation	CHOICE → match method to question purpose

CLASSIFICATION	DIAGNOSTIC BASIS	PROCESS LINK	EXAM TRAP
• KOPPEN → temperature + precipitation thresholds • TREWARTHA → modified thermal zonation	• THORNTHWAITE → water balance + PET • CHOICE → match method to question purpose	• KOPPEN → temperature + precipitation thresholds • TREWARTHA → modified thermal zonation	• THORNTHWAITE → water balance + PET • CHOICE → match method to question purpose

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: THORNTHWAITE → water balance + PET; CHOICE → match method to question purpose.

10

STAGE 10

PLANNING OVERLAY

PLANNING OVERLAY · CLIMATE TYPE · LONG-PERIOD THERMAL-RAIN PATTERN · CROP AND SEASON CONTEXT · SOIL + LANDFORM + GROWING PERIOD · RISK LAYER · HAZARDS, EXPOSURE AND VULNERABILITY

CLIMATE TYPE → long-period thermal-rain pattern

AGRO-CLIMATE → crop and season context → AGRO-ECOLOGY → soil + landform + growing period → RISK LAYER → hazards, exposure and vulnerability

SYSTEM / DEFINITION	PROCESS / MECHANISM	SPATIAL / INDIA PATTERN	HAZARD / LIMIT / RESPONSE
• CLIMATE TYPE → long-period thermal-rain pattern • AGRO-CLIMATE → crop and season context	• AGRO-ECOLOGY → soil + landform + growing period • RISK LAYER → hazards, exposure and vulnerability	• CLIMATE TYPE → long-period thermal-rain pattern • AGRO-CLIMATE → crop and season context	• AGRO-ECOLOGY → soil + landform + growing period • RISK LAYER → hazards, exposure and vulnerability

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AGRO-ECOLOGY → soil + landform + growing period.

11

STAGE 11

CLIMATE ANSWER SPINE

CLIMATE ANSWER SPINE · PURPOSE, VARIABLES AND PERIOD · LETTERS AND THRESHOLDS · INDIA WITH TRANSITION CAVEAT · METHOD, USE, LIMITATION AND VERDICT

DEFINE → purpose, variables and period → DECODE → letters and thresholds → MAP → India with transition caveat → COMPARE → method, use, limitation and verdict

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
• DEFINE → purpose, variables and period • DECODE → letters and thresholds	• MAP → India with transition caveat • COMPARE → method, use, limitation and verdict	• DEFINE → purpose, variables and period • DECODE → letters and thresholds

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MAP → India with transition caveat; COMPARE → method, use, limitation and verdict.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · A CLIMATES IN INDIA INCLUDE TROPICAL HUMID CLIMATES; AM · B CLIMATES ARE DRY · BSHW SEMI-ARID STEPPE AND BWHW HOT DESERT; · CWG IS THE GANGES/NORTH INDIAN PLAINS MONSOON CLIMATE WITH · H/E APPLIES TO THE HIMALAYA ABOVE HIGH ELEVATIONS/SNOW-LINE CONDITIONS. · TREWARTHA MODIFIES KOPPEN AND INCLUDES HIGHLAND CLIMATE FOR INDIA. · THORNTHWAITE USES PRECIPITATION-EVAPOTRANSPIRATION MOISTURE BALANCE, NOT TEMPERATURE ALONE.

OPTIONAL PROCESS DEPTH

- A climates in India include tropical humid climates; Am relates to heavy monsoon rain, while Aw marks savanna/dry-winter interior.
- B climates are dry; BShw semi-arid steppe and BWhw hot desert.
- Cwg is the Ganges/north Indian plains monsoon climate with dry winter.
- H/E applies to the Himalaya above high elevations/snow-line conditions.

DATA / INSTITUTION LIMIT

- Trewartha modifies Koppen and includes highland climate for India.
- Thornthwaite uses precipitation-evapotranspiration moisture balance, not temperature alone.
- India has Koppen A, B, C and H in standard applied schemes.
- Am/Amw = West Coast and very wet monsoon belt; Aw = peninsular savanna interior.

CONTEMPORARY PARALLEL

- As = Tamil Nadu/Coromandel dry-summer tropical type linked to NE monsoon rainfall.
- BWhw = hot desert of western Rajasthan/Thar/Kachchh.
- BShw = semi-arid steppe/rain-shadow and NW margins.
- Cwg = Ganges type: north Indian plains, monsoon with dry winter.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels